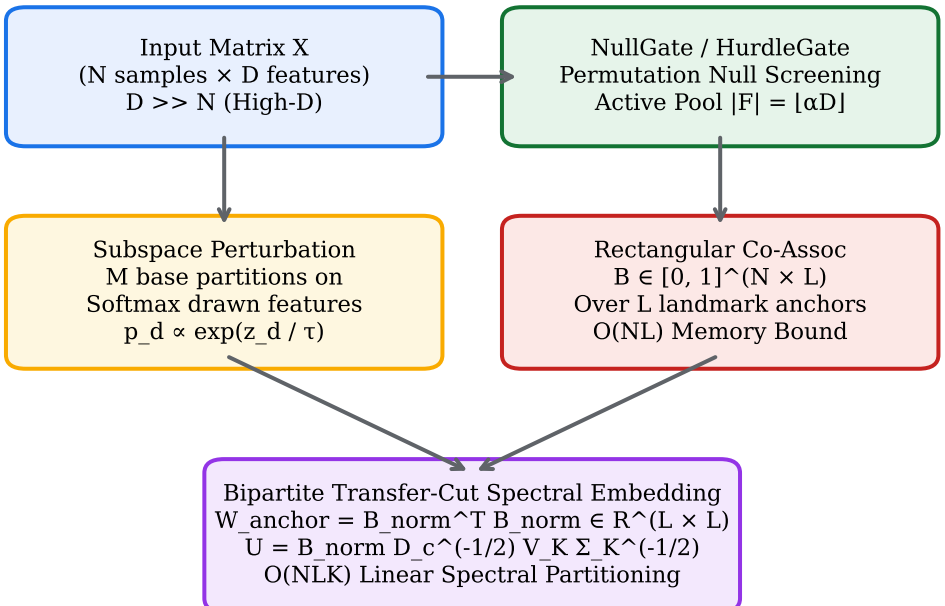
(a) NullConsensus Bipartite Landmark Consensus



(b) Zero-Inflation Dropout Collapse vs. HurdleGate

Standard Feature Gating Failure (Sarle's BC / NullGate):
- Single-cell RNA-seq has 60%–90% dropout zeros.
- Massive point mass at 0 drives kurtosis $\kappa_d \to \infty$, skewness $\gamma_d \to \infty$.
- BC = $(\gamma_d^2 + 1)/(\kappa_d + 3) \to 1.0$ (Artificial Bimodality Spike).
- Selects uninformative noise genes → Catastrophic Collapse (ARI ≈ 0)

Two-Part HurdleGate Decomposition:

$s_d = \alpha \cdot \text{Entropy}(p_{0,d}) + (1 - \alpha) \cdot \text{BC}(x_d^+)$

1. Bernoulli Part: $p_{0,d} = (1/N) \sum 1\{x_{id} = 0\}$
   Measures informative dropout frequency across cellular states.

2. Positive Continuous Support: $x_d^+ = \{x_{id} \mid x_{id} > 0\}$
   Evaluates true biological multimodality strictly on expressing cells.

[+] Eliminates point-mass artifact [+] Restores +0.3832 ARI Recovery